Aadhav Sivakumar

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Education

New York University, Tandon School of Engineering — Brooklyn, NY
Master of Science in Mechatronics and Robotics | GPA 3.95 — *Aug 2024 – May 2026*

- Relevant Coursework: Reinforcement Learning & Optimal Control (MPC, iLQR/DDP, Q-learning, actor-critic), Robot Perception, Interactive Medical Robotics, Advanced Mechatronics, Foundations of Robotics, Entrepreneurship

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science in Robotics Engineering, Minor in Electrical Engineering | GPA 3.8 — *Jun 2020 – Jun 2024*

- Relevant Coursework: Models of Robotic Manipulation, Feedback Control Systems, Microprocessor System Design, Sensing and Sensor Technologies, Advanced Analog Circuits, Signals and Systems, Logic Design, Computer Systems and C Programming, Computer Systems and Assembly Language, Introduction to EDA Tools for PCB Design, Engineering Mechanics: Kinematics and Dynamics, Statics and Dynamics, Introduction to Data Structures and Algorithms, Programming Abstractions: Python

Dublin High School — Dublin, CA
Engineering and Design Academy | GPA 4.2 — *Aug 2016 – Jun 2020*

Experience

AI Engineer — Jan 2026 – Sep 2026
Roboflow | Edge AI Engineering Team — *New York, NY*

- Delivered customer-facing industrial computer vision systems end to end across healthcare manufacturing, food and beverage, poultry, and automotive accounts, covering camera and lens selection, deployment architecture, PLC and controls integration, commissioning, and onsite support
- Served as technical DRI for the Roboflow x Standard Bots robotics integration: built the active-learning loop feeding robot-captured imagery back into training sets, including the dataset upload path for poorly connected sites; root-caused offline and air-gapped inference failures in the model auto-loader and metadata cache; resolved API key, OAuth, and workspace permission blockers; and produced the onsite field kit (architecture and sequence diagrams, verified x86 and Jetson container commands, air-gap warm-up procedure) used for the partner's NYC deployment
- Designed a custom robot control container for the edge deployment stack, letting an arm be driven directly from user-configurable Roboflow Workflows: a sidecar image running alongside the inference and device manager services that maps perception outputs to motion commands, packaged through the field engineering custom container deployment pipeline and demonstrated as closed-loop visual servoing pick-and-place on an xArm 5
- Co-led the Roboflow x NVIDIA synthetic data generation partnership, owning the defect injection, dataset packaging, and train/eval/scorecard stages: ran NVIDIA PAIDF AnomalyGen (few-shot diffusion inpainting on a frozen Cosmos-Predict2 2B DiT with a DINOv2 mask encoder) to synthesize defect data from a handful of real images, established iteration count over dataset size as the dominant training variable, and sourced permissively licensed defect benchmarks
- Authored the internal technical evaluation of NVIDIA's Cosmos 3 omnimodal world model family (architecture, training scale, benchmarks, licensing) and designed the Cosmos 3 Edge on-device evaluation plan for Jetson Thor and DGX Spark, ruling out the larger variants on memory and bandwidth grounds and defining the distill-down deployment path
- Trained and shipped object detection, instance segmentation, and keypoint models for production lines, including an RF-DETR-Seg part-presence model (94.2% mAP@50, 93.8% F1 over 11 classes) tuned through a 560-candidate neural architecture search and optimized with TensorRT from ∼5 FPS to 9.5–11.5 FPS per stream
- Owned edge deployment across the Jetson Orin and Thor lines: device provisioning and BSP/JetPack image work on Advantech, ASUS, and Seeed carriers, a five-container device manager stack (manager, inference, Redis, event store, HMI), Tailscale fleet management, multi-NIC and PoE camera networking, and on-device latency, memory, and power profiling
- Specified a metrology-grade rivet and seam inspection rig for bus roofs in motion (±1 mm over a 1524 mm span): drove the sensor trade study across structured light, ToF, line-scan, and monocular depth, then specified a 16.2 MP LUCID Triton with a 12 mm lens at 0.343 mm/px plus cross-polarized lighting for a ∼$11k build, with hidden-edge inference by least-squares fit through rivet centroids

- Proposed Detection-Grounded Action Models (incomplete; concept stage), an architecture replacing a VLA's vision encoder with a frozen detection, segmentation, and depth stack emitting structured per-object tokens to a language-conditioned action decoder; wrote the v0.1 implementation plan on xArm 5 with GELLO teleoperation and DGX Spark training
- Built and ran the company's booth demos and enablement material: a vision-guided xArm 5 pick-and-place demo for NVIDIA GTC 2026, the CVPR 2026 industrial demo kit, a quality-control webinar, and study guides for the edge stack and JetPack image development

Graduate Robotics Researcher Jan 2026 – Jun 2026

LAIR, NYU Tandon School of Engineering | Advised by Professor Christopher Clark *Brooklyn, NY*

- Developed an Isaac Sim Real2Sim pipeline turning 3D Gaussian Splatting scans of the lab into simulation-ready digital twins, then used scripted planners to generate 1k+ randomized demonstrations at 100% in-sim success
- Helped fine-tune π_0 and $\pi_{0.5}$ flow-matching VLA policies (openpi/JAX) on bimanual Ultra robot LeRobot teleoperation data, benchmarked against an ACT baseline
- Built a bimanual SO-ARM101 teleoperation and data-collection pipeline on Jetson Orin Nano (LeRobot 0.5.2, Feetech STS3215 servos, per-arm calibration, persistent udev device naming)
- Built and deployed a BlueROV2 for 3D reconstruction of underwater shipwrecks, with an Isaac Sim 5.0 and OceanSim simulation environment for mission planning
- Contributed the technical write-up for the lab's Google TPU research award application (openpi, Tunix, vLLM-on-TPU)

Robot Technician Aug 2025 – Jan 2026

Starship Technologies | Fordham University Hub *Bronx, NY*

- Conducted root cause analysis (RCA) on recurring hardware and software faults, developing and implementing long-term solutions that reduced specific error rates and kept up fleet health
- Wrote Python and Bash diagnostic scripts to triage faults in the field, cutting robot downtime by 50%
- Performed component-level replacement of faulty equipment, including circuit boards, Lidar/camera sensors, and power systems, to restore full robot functionality
- Assembled, tested, and calibrated new robotic units to ensure they met operational specifications before deployment into the active fleet
- Calibrated camera and Lidar sensors after repair and re-verified obstacle detection and localization behavior before returning units to autonomous operation
- Performed preventive maintenance and pre-deployment inspections across the hub's active fleet to sustain daily delivery operations
- Documented recurring failure modes and escalated hardware and software issues to the engineering teams with logs and reproduction steps

Robotics Graduate Teaching Assistant Sep 2025 – May 2026

NYU Tandon School of Engineering — Math for Robotics, Foundations of Robotics, Mechatronics *Brooklyn, NY*

- Facilitated weekly office hours and review sessions for 50+ graduate students, clarifying complex concepts in linear algebra, robotic control, and kinematics
- Developed a final project, consisting of a Universal Robots UR10e simulation in MuJoCo with python to test students' understanding of forward/inverse kinematics/dynamics
- Wrote the Mechatronics midterm practical (a multi-mode LED controller on the NUCLEO-H503RB) and updated the course project guidelines for the STM32 platform migration, developing reference implementations for USART printf over ST-LINK, ADC-driven servo control, and ultrasonic and LCD peripherals
- Assisted in grading exams, projects, and homework assignments, providing detailed technical feedback on mathematical proofs and python implementations

Machine Learning and AI Instructor Jul 2025 – Aug 2025

NYU Tandon School of Engineering K-12 — IDEA program *Brooklyn, NY*

- Developed and delivered a comprehensive introductory curriculum on AI and Machine Learning, translating complex topics from classical algorithms to modern neural network architectures for a beginner-level audience
- Instructed students on key machine learning paradigms, including Supervised Learning (Classification, Regression), Unsupervised Learning (Clustering, Association), and the foundational principles of deep learning with neural networks
- Led coding projects using Python, Scikit-learn, and TensorFlow, culminating in a final project where students performed business analytics on real-world Kaggle datasets to derive actionable insights

Electrical Engineering Instructor Jun 2025 – Jul 2025

NYU Tandon School of Engineering K-12 — IeSosc program *Brooklyn, NY*

- Designed and delivered a multifaceted curriculum for NYU's pre-college program, integrating foundational electrical engineering concepts with introductory entrepreneurship principles to provide real-world context
- Led hands-on labs in circuit construction and Arduino programming with hardware like sensors and LCDs; developed and presented a key demonstration measuring Arduino PWM outputs with an analog oscilloscope
- Mentored student teams in developing a functional prototype for their capstone project, culminating in a "Shark Tank"-style presentation where they pitched their invention's technical design and market potential

Undergraduate Research Assistant Jun 2023 – Jun 2024

UCSC Tactile Manipulation Lab *Santa Cruz, CA*

- Worked under professor Tae Myung Huh to develop technologies related to soft robotics
- Engineered a complete sensor solution for a soft robotic end-effector, utilizing a flexible PCB, a Infineon Capsense chip, and custom-molded silicone to detect shear and transverse forces for simultaneous, pressure-sensitive grasping
- Set up lab equipment including laser cutters, Robot arms, 3D and Stereolithography printers, silicone molds, and CUDA for interfacing Ubuntu with an Nvidia RTX 3090 in computer vision applications

LSS Grader and Tutor Sep 2022 – Jun 2024

UCSC Baskin Engineering *Santa Cruz, CA*

- Provided expert tutoring and academic support across a diverse engineering curriculum, including the classes Introduction to Electronic Circuits, Logic Design, and Mechatronics, clarifying complex concepts for over 100 students
- Teaching topics covered: Circuit Analysis and Simulation (MATLAB), Digital Logic Design (Boolean Algebra, Finite-State Machines), Embedded Systems Programming (C), Microcontroller Integration (Sensors, Motors, A/D Converters), and Programmable Logic Devices (PLDs)
- Mentored students by simplifying complex topics, providing constructive feedback on assignments, and developing tailored learning strategies to improve academic performance
- Managed grading schedules and administrative tasks efficiently, ensuring timely and accurate assessment for multiple courses.

Electrical Engineering Instructor Sep 2022 – Dec 2022

UCSC Sustainability Lab *Santa Cruz, CA*

- Created and taught a 3 credit elective electronics course through the university, requiring technical support, project management skills, technical documentation, and procurement of electrical drawings and schematics.
- Lectured to students about topics about the EDA workflow, including soldering and pcb designing/manufacturing.
- Supervised projects that included electrical circuits, specifically the 555 timer circuit and a buck converter circuit.

Learning Technologies Assistant Sep 2021 – Jun 2022

UCSC Learning Support Services *Santa Cruz, CA*

- Maintained and updated technologies within classrooms, including projectors, computers and recording hardware.
- Kept inventory and rented out equipment to organizations and students on campus.
- Assessed and troubleshot computer problems brought by students, faculty and staff.

Stock Data Analyst Intern Jun 2021 – Aug 2021

Foliowiz *Remote*

- Developed and implemented Python scripts utilizing the requests library to automate the extraction of time-sensitive financial data from a third-party API, ensuring an accurate and up-to-date dataset for analysis
- Applied quantitative algorithms using Pandas for data manipulation and NumPy for mathematical operations to analyze stock market data, successfully identifying key performance indicators and forecasting top-performing equities
- Designed and generated client-facing data visualizations and reports with Matplotlib to translate complex analytical findings into clear, actionable insights
- Utilized the Polygon API to build an automated pipeline for ingesting and processing comprehensive market data.

Code Coach Jul 2019 – Jun 2021

theCoderSchool *San Ramon, CA*

- Mentored over 30 students (ages 7-18) in core programming principles, utilizing languages such as Python, Lua, and Scratch to guide them from basic syntax to fully functional project completion
- Translated complex computer science concepts, such as object-oriented programming and data structures, into digestible, age-appropriate lessons, fostering critical thinking and debugging skills in 1-on-1 coaching environments
- Analyzed and documented student progress through weekly reports to identify knowledge gaps, enabling data-driven adjustments to lesson plans for improved concept mastery

Projects

Real2Sim2Real SO-ARM101 | *Personal Project, Isaac Sim/Lab, gsplat, COLMAP, LeRobot, ACT* Sep 2026 – Oct 2026

- Taught a low-cost SO-ARM101 arm to place a 30 mm cube in a cup by co-training an ACT policy on 40 real teleoperated demonstrations and about 4,000 simulated ones, reaching 19/21 real-world successes versus 6/21 for real-only baselines
- Reconstructed the desk from a 45-second phone video with COLMAP and gsplat (about 347k Gaussians) and used the splat as the background of an Isaac Sim scene aligned to the real cameras through ChArUco lens calibration, overhead camera pose, joint zero offsets, and wrist-mount fitting
- Generated randomized demonstrations with a scripted expert in Isaac Lab using domain randomization, DART-style action noise, and DAgger-style recovery takeovers, then audited the sim data and removed 1,509 episodes that spawned off the real desk
- Targeted simulated data at gaps in the real demonstrations (336 sim episodes vs. 4 real demos near the real-only failure region), weighted real data to about 25% of frames in the co-training mix, and reached 88% success on 50 held-out sim scenes
- Deployed on a Jetson Orin Nano Super with fp16 and CUDA graphs (about 40 ms per action chunk vs. 72 ms in fp32) behind a web console that labels and logs every run; also experimented with SmolVLA and GR00T N1.7

Detection-Grounded Action Models (DGAM) | *Roboflow R&D* | **INCOMPLETE** Jul 2026 – Sep 2026

- Architecture concept replacing a VLA's learned vision encoder with a frozen CV stack (detector, instance segmenter, depth) that emits structured per-object tokens: class, mask shape, 3D pose, affordance keypoints, and spatial relations
- Tokens are consumed by a lightweight language-conditioned action decoder, trading end-to-end pixel learning for interpretable, swappable perception
- Surveyed related approaches (ObjectVLA, Oat-VLA, SVLR, VoxPoser, SayCan, Point Policy, Afford2Act) to position the design against existing object-centric and language-conditioned policies
- v0.1 plan: xArm 5 with wrist and scene cameras, RF-DETR detector, GELLO teleoperation, ACT-style chunked actions (chunk 16, L1 loss), 4-task pick-and-place domain in LeRobot dataset format on DGX Spark

Part Presence Inspection System | *Roboflow Webinar Demo* Aug 2026 – Sep 2026

- Built a two-camera inspection cell (LUCID Triton and Basler ace 2 on a Jetson) verifying component presence in an automotive intake manifold and carburetor kit, alerting through a custom HMI
- Trained an RF-DETR-Seg model across 11 classes on 536 labeled images and 6,063 annotations, augmented to 1,238, reaching 94.2% mAP@50, 94.2% precision, and 93.4% recall
- Raised throughput from ∼5 FPS to 9.5–11.5 FPS per stream with TensorRT, a 432×432 input, sensor-side ROI cropping, and cheaper mask visualization
- Designed the supporting bench hardware: a single-axis belt-drive motion stage, telecentric lens selection with a coverage calculator, and coaxial lighting

Rivet and Seam Metrology Rig | *Roboflow Customer Engineering* Feb 2026 – Jun 2026

- Measurement system for 5 mm rivets on the overlapping seam of curved bus roof sheets, ±1 mm accuracy over a 1524 mm span with the vehicle moving at 3–5 mph
- Ran the sensor trade study (RealSense and Photoneo structured light, Helios2 and blaze ToF, blue laser profilers, line-scan with encoder trigger, Depth Anything 3 metric depth) and ruled out monocular and ToF depth for mm-scale metrology
- Final spec: LUCID Triton TRI162S (16.2 MP, IMX542) with a Fujinon 12 mm f/1.8 at 1.5 m working distance for 0.343 mm/px, cross-polarized LED bars, and an optional ToF unit for depth correction, at ∼$11,150
- Accuracy plan used rivets as a reference scale (0.75 mm to 0.03 mm scale error), ∼100-point averaging per seam, and depth-aware projection; hidden bottom-sheet edge inferred by least-squares line fit through rivet centroids

Machine Vision Lens Calculator | *Internal Tool* Apr 2026 – Jun 2026

- Single-file HTML/JS tool (∼1,500 lines) for specifying industrial camera and lens pairs in the field
- Computes exact thin-lens and angular FOV, magnification, working f-number with pupil magnification, hyperfocal distance, near/far depth of field, resolution at the FOV plane, and Airy disk diameter
- Cascading selector across the LUCID Triton, Triton2, Atlas, and Basler ace 2 lineups, with inverse solves for focal length and working distance and SVG side and front view diagrams

GELLO Leader Arm for xArm 5 | *Teleoperation Hardware* May 2026 – Aug 2026

- Modified, printed, and assembled a GELLO-style leader arm for collecting VLA demonstration data on an xArm 5

- Dynamixel XL330-M288-T joint servos and an XL330-M077-T gripper on FPX330 frame kits, driven from a 5 V 4 A supply
- Mapped xArm 7 reference STLs onto xArm 5 kinematics, verifying that encoder readings transfer 1:1 despite the link mismatch

Synthetic Defect Generation with Cosmos and AnomalyGen | *NVIDIA Partnership* Jun 2026 – Sep 2026

- Evaluated NVIDIA's PAIDF AnomalyGen, a few-shot diffusion inpainter with ∼3 M trainable adapter parameters on a frozen Cosmos-Predict2 2B DiT with a DINOv2 mask encoder, for generating defect data from a handful of real examples
- Ran training experiments establishing iteration count rather than dataset size as the dominant variable, and characterized where mask geometry causes failures
- Sourced benchmark datasets under permissive licenses (DeepPCB, VisA, filtered Roboflow Universe) after ruling out non-commercial industrial anomaly sets
- Assessed GPU feasibility and deployment of the generation webapp on H100-class hosts, and documented an internal experiment platform as the path for scaled runs

Vision-Guided xArm 5 Pick-and-Place | *NVIDIA GTC 2026 Booth Demo* Feb 2026 – Mar 2026

- Built a visually servoed pick-and-place demo sorting bottles with an xArm 5, a RealSense D435i, and a Roboflow Workflows inference server
- Ported a Blockly program to the xArm Python SDK, then added closed-loop visual servoing with a tuned proportional gain, camera-to-gripper offset calibration, time-based descent, and keyboard emergency stop
- Brought the full stack up on an Advantech MIC-733AO (JetPack 6.2.1, ARM64): Jetson inference container, PoE Basler camera on a link-local subnet, RealSense as a V4L2 device after SDK backend failures, and direct arm-to-Jetson networking
- Configured the end-effector payload profile (1.056 kg with gripper, camera mount, and depth camera; 177 mm TCP offset) and validated air travel logistics for the booth

UR10e Mujoco Simulation | *TA-created Final Class Project* Sep 2025 – May 2026

- Created as a final project for the students as a TA for Foundations of Robotics
- Uses mujoco to handle control and collisions
- Requires inverse kinematics and dynamics calculations to control robot to move to desired point, or move at a desired velocity

3D SLAM Drone with Autonomous Landing | *Master's Project* | **INCOMPLETE** Sep 2025 – Dec 2025

- Began building a Pixhawk drone for Project Millet to test autonomous pesticide spraying across an orchard
- Designed a system pairing a UGV carrying a liquid payload with a drone fitted with a high-powered sprayer
- Started a small-scale drone for planning alongside a large-scale drone for field experiments
- Planned perception and path planning on an Nvidia Jetson with an Intel RealSense depth camera

Unitree Go2 RL Locomotion | *Reinforcement Learning & Optimal Control Class Project* Nov 2025 – Dec 2025

- Trained a quadruped locomotion policy for the Unitree Go2 with PPO (RSL-RL) in Isaac Lab, running training jobs on NYU's GPU cluster through Slurm and Singularity, as part of a three-person team
- Replaced the default actuation with a custom PD torque controller with tuned gains and torque limits, and added early termination for episodes where the base dropped below a height threshold
- Structured the gait with a Raibert heuristic for foot placement and phase timing
- Shaped the reward to penalize action rate, body tilt, vertical bouncing, and high joint velocities, and to reward foot clearance and ground contact forces

SoleGait Foot Sensor | *NYU Vertically Integrated Project* Jan 2025 – Present

- Developed a smart shoe sole that track the gait and pressure of your foot as you walk or run
- Integrates force sensors with an Arduino Uno, transmitting the force and gyroscope data over UART to a computer to display real time data on MATLAB
- Ported the sensing node to a Seeed XIAO nRF54L15 Sense running Zephyr under the nRF Connect SDK, streaming 104 Hz IMU data to a MATLAB viewer with a complementary filter for real-time 3D attitude
- Won final prize for best design at the NYU 2025 capstone competition

Glass-2-Bot | *Advanced Mechatronics Final Project* Apr 2025 – May 2025

- Completed as the last mini-project of the Advanced Mechatronics course at NYU

- Engineered a hands-free user interface by streaming 720p video from a depreciated Google Glass Explorer Edition to a Raspberry Pi, enabling real-time item selection via object detection
- Built a mobile manipulator by modifying and 3D-printing an open-source design and developed a dual-microcontroller architecture where a Raspberry Pi handles computer vision and a C++ programmed Arduino executes a state machine for autonomous navigation and grasping

2R Manipulator Sand Table | *Advanced Mechatronics Project 2* Mar 2025 – Apr 2025

- Completed as the second mini-project of the Advanced Mechatronics course at NYU
- Engineered a low-cost kinetic sand art table by fabricating a 3D-printed 2R manipulator and programming a C-based state machine on a Parallax Propeller to translate joystick inputs into stepper motor commands via inverse kinematics, enabling user-drawn and autonomous patterns
- Designed the project's physical assembly in Fusion 360 and simulated the manipulator's motion in MATLAB before fabricating the 3D-printed arm links and wooden housing

Arduino PONG | *Advanced Mechatronics Project 1* Jan 2025 – Feb 2025

- Completed as the first mini-project of the Advanced Mechatronics course at NYU
- Engineered a portable Pong game by integrating hardware with an Arduino Mega and developing optimized code to resolve complex LED matrix mapping and memory-usage constraints
- Designed and manufactured a custom game console chassis and ergonomic controllers using Fusion 360, laser cutting, and 3D printing while managing the complete 3-week project lifecycle from validation to final testing

ASL Glove Interpreter | *Mechatronics Final Project* Nov 2024 – Dec 2024

- Created for the final project of the Mechatronics class at NYU
- Utilized multiple flex sensors, an LCD screen, and the Basic Stamp 2, for inputs, outputs, and processing
- Was able to correctly identify all numbers, input multiple digit numbers, and do basic mathematical operations

SMART Compost Sorting | *Senior Capstone Project* Sep 2023 – May 2024

- Created a solution for the problem of contaminated compost streams on UC Santa Cruz campus
- Used OpenCV with DexNet to determine compost versus contaminant and remove it with a robot arm
- Utilized 3D space mapping with transformation matrices to map the depth camera output to robot joint space

Stockbot: Grocery Manipulator Simulation | *ECE215 Final Project* Feb 2024 – Mar 2024

- Completed as part of ECE215 (Models of Robotic Manipulation)
- Developed and implemented a PID-based feedback control system in each joint of a 7-Degree-of-Freedom robotic arm (Franka Panda) to create a multi-item pick-and-place testing environment
- Instrumented the robotic workcell to log and analyze real-time performance telemetry, including cycle times and task success rates, creating a validation framework to benchmark system efficiency against human trials

IMU driver calibration | *ECE167 Final Project* Feb 2024 – Mar 2024

- Completed as part of ECE167 (Sensors and Systems)
- Developed an embedded C-based AHRS, fusing 9-DOF IMU data with a complementary filter and propagating attitude via a Direction Cosine Matrix (DCM)
- Performed full IMU calibration (gyro bias, axis misalignment) and tuned complementary filter gains after validating the model in MATLAB

PCB Design for milling | *ECE174 Final Project* Nov 2023 – Dec 2023

- Completed as part of ECE174 (Intro to EDA Tools)
- Learned about all of the intricate design philosophies involved with PCB layout and fabrication
- Created an audio amplifier out of base components and milled PCB

Tactile Manipulation Sensor | Research Project Jul 2023 – Jun 2024

- Developed as part of research at the UCSC Tactile Manipulation Lab
- Designed and developed a sensor for a soft gripper that can be mounted on the end effector of a robotic arm, allowing objects to be sensed and grabbed simultaneously using varying pressures
- Sensor consists of flexible PCB, cypress Capsense chip, and custom molded and cured silicone

Mechatronics Competition | *Undergrad Mechatronics* Apr 2023 – Jun 2023

- Final Project as part of Mechatronics course at UCSC

- Involved fully prototyping an ping-pong ball shooting robot in Solidworks, and manufacturing it with a laser cutter and power tools
- Competed against other teams within the class, aand got the most points in a single round

FPGA VGA Game | *CSE100 Project* May 2022 – Jun 2022

- Completed as part of a lab in CSE100 (Logic Design) at santa cruz
- Used the Digilent Basys 3 board, programmed with verilog
- utilized manual recreation of VGA protocol, while processing inputs to create a fully functional Flappy bird-like game

MATE ROV Competition Robot | *Slugbotics* Sep 2021 – May 2023

- Created PCBs and other electrical circuits to control the movement and sensors of an underwater drone
- Designed CAD model for the inside of the waterproof enclosure housing the main electronics
- Coordinated troubleshooting and maintenance of power equipment

VEX Robotics | *High School Robotics Competition* Aug 2016 – May 2020

- Served as captain of VEX robotics team 5327B
- Created competitive robots that of competed at the regional, state, and world level
- Learned concepts like PID control, slew rates, and CAD for complex constructions
- Incorporated complex design processes, like CAD, simulation, decision matrices, and iterative design

Automated Dog Feeder | *Side Project* Jun 2016 – Aug 2016

- My first real project, completed with my dad
- Created a system that dispenses the standard amount of dog food in a bowl
- Connected with IFTTT to activate remotely based on sending an email to a private account
- Uses a simple DC motor, a cereal dispenser, and a drum of dog food to be refilled manually for dispensing

Technical Skills

Languages: C, C++, Python, Micropython, MATLAB, Verilog, Java, HTML/CSS, Javascript, SQL
Frameworks: ROS2, PyTorch, Tensorflow, JAX, LeRobot, openpi, RSL-RL, TensorRT, OpenCV, Open3D, gsplat, COLMAP, numpy, pandas, scikit
Models: RF-DETR, YOLO, SAM2, DINOv2, $\pi_0/\pi_{0.5}$, ACT, SmolVLA, GR00T N1.7, Cosmos (Predict2, Cosmos 3), PAIDF AnomalyGen, Depth Anything
Edge & Infrastructure: NVIDIA Jetson (Orin, Thor, AGX/NX), JetPack/L4T, DGX Spark, Docker, Redis, TimescaleDB, Grafana/Loki, Tailscale, Linux/Ubuntu
Boards: Arduino, Raspberry Pi, Digilent Basys 3 FPGA, STM32 (NUCLEO-H503RB), ESP32, nRF54L15, Parallax Propeller, BASIC Stamp 2
Hardware: Franka Research 3, UFactory xArm 5, SO-ARM101, BlueROV2, Dynamixel/Feetech servos, NI DAQ, Bambu Lab printer, Glowforge
Machine Vision: GigE Vision/USB3 Vision, LUCID (Triton, Helios), Basler ace 2, Intel RealSense, Orbbec, structured light and ToF, lens and lighting selection, calibration (ChArUco/ArUco)
Software: Solidworks, Altium, EAGLE, Fusion 360, Cadence, Mujoco, Isaac Sim/Isaac Lab, WeBots, PSpice, AutoCAD
Robot Learning: VLA fine-tuning, imitation learning, reinforcement learning (PPO), reward shaping, action chunking, leader-follower teleoperation, LeRobot/RLDS datasets, Real2Sim, sim-to-real, sim/real co-training, domain randomization, DAgger-style data aggregation, synthetic data generation
Other tools: Communication protocols (SPI, I2C, UART), Inverse kinematics, visual servoing, hand-eye calibration, PID control, Signal Filtering, Driver Creation/Calibration, Statics and Dynamics, ATI multi-axis force/torque, 9-axis IMU